Personalizing and Optimizing Learning Experiences using AI

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Abstract

With the rapid expansion of online educational resources, learners often face the challenge of discovering content that aligns with their goals, interests, and prior knowledge. Using AI's powerful machine learning features alongside a course catalog such as Microsoft Learn Course Catalog open API, we can create a powerful learning experience that is personalized for each user. Enhancing the learning experience for each individual user by having the AI give useful recommendations for courses that are related to or that the user may have interest in is a powerful feature. Users may continue their learning path based on their experiences which may result in a better learning path than by using linear and more traditional methods. This paper aims to bring a collaborative approach to understand and determine AI's use in the future of education. Diving into strategies that are meant to take user's suggestions and actions to provide recommendations that will lead these users to partake in unique learning experiences. Our desired outcome is to provide the educator and students with arguments and data that may lead them to see the possibilities of personalized learning solutions for groups and individuals.

1 Introduction

The rise in massive open online courses has transformed how we access knowledge, and have provided an abundant amount of learning resources throughout all disciplines. However, traditional learning platforms often use one size fits all approach which makes it very difficult to navigate through all available content that cater to our needs. Artificial Intelligence (AI) and machine learning advancements have opened up the possibilities to achieve gigantic strides in personalized learning environments. The use of AI in learning platforms provides dynamic, customized recommendations based on learner interests, behavior, and preferences. Linear course structures cannot provide for diversified user interests and learning behaviors as presented in traditional systems, making e-learning inadequate. Relying on machine learning algorithms paired with deep course catalog APIs, like the Microsoft Learn Course Catalog open API, is the secret to delivering adaptive, personalized learning experiences. This study investigates if AI-powered recommendation systems can be utilized in a bid to enhance user engagement and learning achievement by exposing users to pertinent course suggestions that map to individual learning trajectories.

We will tackle the challenges and possibilities that come with using an AI assisted tool with an API and measure the effectiveness of these tools. We will examine how AI, Machine Learning, deep learning models and algorithms may create custom educational materials in a broad amount of subjects [17]. We will also look into how using these technologies to focus on adaptive education and to bring personalized approaches will further adapt learning features [15]. This paper will deal with the possibilities to help populations at risk by arguing the enhancement of infrastructures that may bring these technologies to reality in these communities [12]. Overall focusing on the long term use of this technology and how it may eventually help enhance our education system, but to do so we have to continue to grow our technological infrastructure in order for greater reach and accessibility [13].

We will look into how personalization and recommendation systems have been studied so far in education as well as other fields and try to incorporate it in our project. Several projects and papers have been published that discusses the use of Artificial Intelligence driven recognition systems such as [8] which takes sentiments of customer reviews into account and makes the model learn the behavior pattern to recommend more useful content. Similarly, [4] utilizes students learning preferences to to dynamically recommend new content. There has also been studies that talk about recommendation system algorithms such as collaborative filtering and content filtering [6] and also discusses the challenges such as cold-star and data sparsity.

Our system will incorporate Content-based filtering and other machine learning models to understand student's goals and learning preferences. We aim to improve our users educational growth by providing most relevant Microsoft courses. We will also discuss methodology, implementation and evaluation in the next parts.

This project explores the application of Natural Language Processing (NLP) to build a semantic course recommendation system that personalizes the educational course discovery process. The system focuses on Microsoft Learn as the content source which is a platform known for its professional-grade, modular courses in cloud computing, AI, software development, and more. The system takes in a user prompt which is phrased naturally rather than exact keywords indicating what they want to learn about and uses semantic understanding techniques and recommends a most relevant list of ten courses in a pagination format that best match the underlying intent of the user. This project utilizes TF-IDF (Term Frequency–Inverse Document Frequency) vectorization combined with cosine similarity, which enables the extraction of semantically similar content based on textual features.

The recommendation system also includes two additional innovations: course summarization and relevance filtering. The summarization component provides a concise and accurate overview of lengthy course descriptions into clear, brief summaries to help in faster comprehension. Meanwhile, the filtering mechanism ensures that the recommended courses do not include duplicates or overly similar content, improving the diversity and usability of suggestions. These features collectively create a more engaging and streamlined user experience.

Built entirely in Python, the tool interfaces directly with the Microsoft Learn Catalog API which has open access, ensuring that course data remains up to date without manual intervention. The recommender supports real-time user inputs and dynamically generates results, making it suitable for both academic advisors and self-directed learners. The project offers a practical demonstration of how semantic search and summarization can make online learning more intuitive, effective, and user-centered.

2 Related Works

- Rane and Choudhary [15] Discusses methods of education and discuss how integrating AI for a personalized approach should be the next step for our education system, labeling it as

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education 5.0. They discuss adaptive learning strategies and how to implement them using AI with tools such as machine language, natural language and image processing. The paper also deals with controversial and ethical issues that come with working with AI such as privacy and the handling of data.

- Adenubi and Samuel [12] Examines how AI can be used to tackle specific education needs and focus on the possibilities it may bring to communities especially ones in vulnerable populations. The study focuses on at-risk students and provides ways in which AI analytics and machine learning may help these students with their learning outcomes. They make a stance on how integrating these strategies may be challenging at first but with the proper resources and training for students and educators on how to use these tools will be beneficial in the long run.

- Sundari, Penthala and Nayyar [17] Presents a study on innovating uses for AI for education with a focus on personalization and content creation. They focus on AI curating and customizing educational materials and tools to enhance the effectiveness of educational content. Using an intelligent recommender system as they call it, a user may search for a specific topic and get various recommended materials from presentation, powerpoints, research documents and tools all based on the user's preference. Raw information data from the user's preference may then identify patterns for the AI system to keep recommended specific materials.

- Nabhani, Hamzah, and Abuhassna [13] Investigates the use of AI in education to improve academic performance, and challenges traditional teaching methods. The research investigates a sample group made of forty ninth grade students and forty educators in which the students were all given training material based on the teacher's input and recommendations. Using questionnaires the sample groups gave feedback on the expectations and experiences from this investigation, resulting in the conclusion where both agree that using AI tools provide an effective and personalized experience that should continue to develop with the academic programs.

- Dhananjaya [5] Analyzes how recommendation systems that use AI can improve engagement and focus on the specific needs for students. This paper focuses on indicators such as goals, motivators, time limits to engage with students and also obtain data that may be used to amplify the performance of the AI tools. It studies the importance of group-based educational recommendation and personalized learning features that combined may result in optimized learning experiences.

- [6] discusses an in-depth examination of Collaborative Filtering method for recommendation systems in spaces like e-commerce. The authors have discussed various recommendation methodologies such as content-based filtering, collaborative filtering as well as hybrid methods. They show how these methods rely on user behavior and preferences and may produce challenges such as data sparsity for new users, cold-star for new items in the market and scalability issues. They say addressing these challenges are crucial for future recommendation systems.

- [8] proposes a different approach for recommendation systems by integrating sentimental analysis from textual reviews with collaborative filtering. They aim to leverage ensemble learning methods to provide more relevant recommendations. They used Bi-LSTM model for sentiment analysis and unsupervised "GloVe" vectorization for classification. This captures users experience and reviews accurately and the result is used integrate with Collaborative Filtering for better recommendations.

- [7] is a comparison between two most used approaches for recommendation, Collaborative Filtering and Content-based Filtering. This paper discusses similarities and challenges in both. According to this paper, Content-based Filtering is majorly used to generate recommendations based on users past utilized content. However, this approach may fall short in recommending content outside of user's domain. On the other side, Collaborative Filtering suggests items and content based on what other users have consumed and liked. It analyzes behavior patterns and introduces new recommendations outside of users domain but may introduce challenges like data sparsity and cold star.

- [4] explores the application of Artificial Intelligence in educational growth of a student. This paper determines influential factors that affect students success in higher education, forecast student's status like dropout or graduation and develop a recommendation system. It discusses various algorithms for personalization such as SVM, Logistic Regression, Random Forest, CatBoost and XGBoost out which Random Forest proved to be the best across all evaluation metrics. It interprets that AI proves to be highly influential in educational success with these AI driven recommendations.

- [11] This paper takes a different approach for the recommendation system for student growth in education. This recommender utilizes key skills required for jobs in a specific field and recommends content that fulfills those key requirements. It uses NLP and text mining techniques to extract relevant skills, maps it to Open Educational Resources and then ranks the educational material with content-based filtering. Current skill of the user and difficulty level of the courses are taken into consideration in recommendations. In their result section, 82.8% of the recommendations were deemed useful.

- [9] Education recommender systems most often apply collaborative filtering, which predicts user interests based on similar users' behaviors, and content-based filtering, which employs content metadata to suggest relevant courses. Both are powerful in recommending learning material but can be vulnerable to cold-start or data sparsity situations. They, therefore, provide a foundation upon which more sophisticated recommendation techniques are developed.

- [3]To overcome drawbacks in stand-alone filtering techniques, hybrid recommendation systems integrate collaborative and content-based filtering, which maximizes flexibility and precision. Hybrid systems utilize complementary advantages to reduce weaknesses of individual techniques. They have been successful on educational platforms to enhance individualized learning experiences and system performance overall.

- [10] Conversational recommenders carry out interactive conversations with learners in order to update recommendations dynamically based on immediate user feedback and preferences. This adaptive style of conversation significantly exploits personalization by allowing the system to clarify user needs on a continual basis. These interactive methods have shown promise in increasing learner satisfaction and engagement.

- [19] NLP methods browse course descriptions and student-generated content to improve recommendation precision. By using the automatic extraction of relevant semantic features

from textual content, such systems are able to match user interests more accurately with offered courses. NLP-driven recommendation systems thus effectively improve relevance and quality of suggested learning material.

- [16] User behavior analytics is significant in enhancing recommendation flexibility through the analysis of fine-grained interaction patterns in online learning platforms. Early interactions in courses consistently predict future learner engagement, making it feasible to proactively tighten up recommended learning pathways. Monitoring learner behaviors allows learning platforms to better anticipate and serve individual student needs.

- [1] Latent pattern analysis of engagement patterns identifies underlying influences on learner persistence and participation in online learning. By tracking these latent patterns, recommendation systems can more accurately forecast learning outcomes and learning preferences of learners. With this type of knowledge, one can make informed recommendations for maintaining learner motivation or stimulating motivation.

- [14] Using large datasets from intelligent tutoring systems enables detailed analysis of student interactions to offer highly personalized recommendations. Methods like Hidden Markov Models and decision trees use fine-grained student and skill data to make better predictions and recommendations. This work successfully personalizes the delivery of educational content based on individual student progress and performance.

- [18] Research in Massive Open Online Course (MOOC) design highlights the importance of adaptive testing strategies and personalization of content delivery to promote learner experience. Effective course design integrates personalized recommendation methods to increase learner retention, satisfaction, and overall performance. As MOOCs are increasing in popularity, it becomes more important to optimize their instructional design through AI-based personalization.

- [2] Examine clickstream data from websites like edX and Coursera, giving insightful information about learner behavior and its contribution to performance estimates. Learner engagement through clickstream data analysis enhances course attendance prediction and dropout risk modeling. Clickstream analysis-based suggestions significantly enhance course personalization and student performance.

3 METHOD

The implementation of this program is divided into three parts which are meant to fetch the data from the API, filter pulled data and display relevant results based on the parameters given. The data is pulled from the Microsoft Learn API which is an open course catalog that allows the program to make a search using a command prompt obtained by the user. This pulled data is then divided into program segments that will include a program description that summarizes five courses that the algorithm selects, which includes a base course and four related courses. This base course and additional related courses are all displayed with a sentence description of the course followed by the URL. The program will display ten of these program segments at a time and gives the user an option to display more results or exit the program once they obtain the wanted information.

The filtering of data contains a mixture of NLP tools that begin by segmenting text into sentences and into words, then calculating text similarity based on their semantic content. This process converts the content to a TF-IDF vector and then uses a ranking system to determine the similarity to the user's query. This ranking system works to determine how semantically close the user's query is to the course description and then it goes further by acquiring similar course information. Once the filtering is completed a coherent paragraph containing a summary of the program with all its courses is presented to the user using sentence tokenization.

The technologies and libraries in use to make this program include an HTTP GET request to fetch all data from the Microsoft Learn servers using its public API at "https://learn.microsoft.com/api/catalog". The scikit-learn library uses a text vectorization method that converts text into numerical vectors using TF-IDF which translates to Term Frequency-Inverse Document Frequency. Once the text is vectorized it looks for cosine similarity between these vectors using a score provided which in this program is using a threshold of 0.05. The data pulled by this request is then parsed into the Natural Language Toolkit (NLTK), which is a library that is used to work on language data. The Punkt sentence tokenizer is what is used to divide text into a list of sentences and then put together these sentences using unsupervised machine learning. The tokenizer figures out concise sentences to form understandable descriptions for the course program and the courses which make up this program by taking care of correct punctuation, splitting sentences, adding correct abbreviations and detecting frequent word and sentence combinations to figure out the start and the end of each description.

3.1 Example

After running the program the user will be asked to enter a topic to search in Microsoft Learn and it will follow by creating ten results all with program descriptions, courses and links as follows:

Enter a topic to search in Microsoft Learn: cloud gaming

Searching Microsoft Learn...

Fetched 4414 total items from MS Learn catalog.

Ranking results for relevance to 'Cloud gaming'...

Found 241 relevant items (threshold: 0.05)..

Top Relevant Items with Relevancy Score, 0 for no relevancy to 1 being identical shown in decimal

Score: 0.5482 - Program Segment: Introduction to gaming and disability

Score: 0.3206 - Program Segment: Accessibility best practices for gaming hardware

Score: 0.3070 - Program Segment: Gaming assistive technologies

Score: 0.2804 - Program Segment: Gaming accessibility fundamentals

Score: 0.2675 - Program Segment: Accelerate cloud adoption with the Microsoft Cloud Adoption Framework for Azure

Score: 0.2217 - Program Segment: Secure your cloud apps and services with Microsoft Defender for Cloud Apps

Score: 0.2187 - Program Segment: Explain cloud workload protections in Microsoft Defender for Cloud

Score: 0.2182 - Program Segment: Plan for cloud workload protections using Microsoft Defender for Cloud

Score: 0.2081 - Program Segment: Microsoft Cloud for Manufacturing overview

Score: 0.2064 - Program Segment: Get started with Microsoft Cloud for Healthcare

Displaying Microsoft Learn Program Segments 1 to 10

Program Segment 1

Analysing related courses...

Course Program Overview

This module introduces accessibility concepts for game titles, gaming platforms, and gaming hardware. You learn about the core elements in gaming experiences that influence accessibility and how those elements can affect the experiences of gamers with visual, hearing, mobility, and cognitive disabilities. At Team Xbox, we believe that when everyone plays, we all win. We're committed to ensuring as many developers as possible have the tools needed to create accessible experiences. This learning path introduces foundational concepts about gaming accessibility for game, platform, and hardware developers looking to expand their knowledge in this space. Learn about the types of assistive hardware and software technologies that players with disabilities use to game, and the importance of supporting these assistive technologies when you create accessible gaming products and experiences for all audiences.

Included Resources:

Core Course: Introduction to gaming and disability Description: This module introduces accessibility concepts for game titles, gaming platforms, and gaming hardware. You learn about the core elements in gaming experiences that influence accessibility and how those elements can affect the experiences of gamers with visual, hearing, mobility, and cognitive disabilities. https://learn.microsoft.com/en-us/training/modules/intro-to-gaming/?WT.mc$_id = api_Catalog Api$

Related Courses (4):

- Gaming accessibility fundamentals Description: At Team Xbox, we believe that when everyone plays, we all win. We're committed to ensuring as many developers as possible have the tools needed to create accessible experiences. This learning path introduces foundational concepts about gaming accessibility for game, platform, and hardware developers looking to expand their knowledge in this space. https://learn.microsoft.com/en-us/training/paths/gaming-accessibility-fundamentals/?WT.mc$_id = api_Catalog Api$
- Gaming assistive technologies Description: Learn about the types of assistive hardware and software technologies that players with disabilities use to game, and the importance of supporting these assistive technologies when you create accessible gaming products and experiences for all audiences. https://learn.microsoft.com/en-us/training/modules/gaming-assistive-technologies/?WT.mc$_id = api_Catalog Api$
- Accessibility best practices for gaming hardware Description: Learn how to create more accessible standard gaming hardware products, such as game consoles, controllers, and headsets. Discover ways to design hardware packaging that features accessible components to improve the unboxing experience of more customers. https://learn.microsoft.com/en-us/training/modules/hardware-devices/?WT.mc$_id = api_Catalog Api$
- Community collaboration for accessible design Description: Learn about the importance of collaboration with the disability community in creating accessible gaming experiences. Identify methods to build connections with these communities. Implement best practice etiquette during these collaboration sessions. https://learn.microsoft.com/en-us/training/modules/community-collaboration/?WT.mc$_id = api_Catalog Api$

4 Results

To quantify the performance of our recommendation system based on AI, we conducted a user test with 16 subjects who were randomly selected from Smyrna, Georgia. The subjects were 9 females and 7 males with varied educational backgrounds and professional fields of work to generate a randomized pool of user experiences. Participants were given access to our course recommendation platform with Microsoft Learn Course Catalog API, augmented with TF-IDF vectorization, and summarization module. They were asked to input natural language queries relating to their learning interests or choice of learning pathway and explore top course recommendations curated by the system. 81% (13 of 16) reported that the summarized course descriptions made it far easier to identify the relevance of each recommendation. Most participants could identify at least two highly relevant courses on the first page of results. Participants who were less technically proficient were most assisted by the natural language input and summarized results, proving that the system is most valuable for non-specialists. Also, Figures 1 and 2 shows the relevance score for each input into the program, referencing how similar each input is to the top relevant items provided.

ID#	Input Type	Top Course Title	Relevancy Score
1	Microsoft Teams	"Facilitate meetings and events with Microsoft Teams"	0.16
2	Building Surveys	"Create surveys with Dynamics 365 Customer Voice."	0.42
3	Slide Design, Presentation, and Animation	"Build effective presentations with AI."	0.19
4	Management, Non-profit, Organization	"Manage and engage with volunteers in Microsoft Cloud for Nonprofit "	0.36
5	Education & Minecraft	"Minecraft Trainer Academy: Train educators to teach with Minecraft Education in the classroom."	0.30
6	Personalization & Windows 11	"Engage teachers and students with Windows 11."	0.22
7	Microsoft Bookings & Schedule Appointments	"Automate scheduling with Microsoft Bookings"	0.29
8	*Unsure, Quality Assurance	"Introduction to AI for business users"	0.16

Figure 1: Data from queries conducted

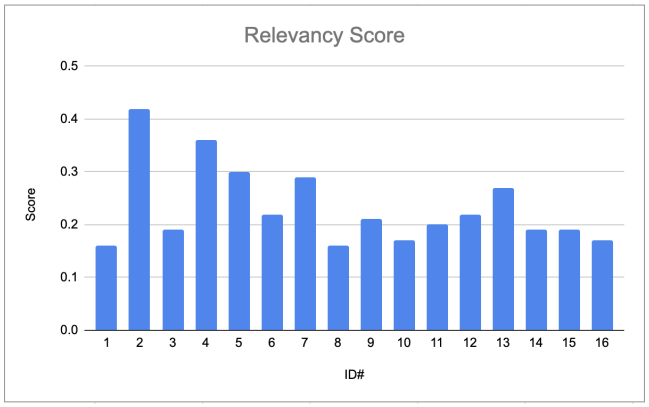

Figure 2: Relevance Score

5 Discussion

The online course catalogs like Microsoft learn when integrated with Natural Language Processing techniques can significantly personalize educational experiences. This project is lightweight, accessible and effectively recommends courses by using AI which enables users to utilize Microsoft Learn resources successfully. Direct keyword searches has limitations when users use natural language phrases or conceptual prompts to search for courses. So, by utilizing TF-IDF vectorization and cosine similarity, our recommendation system performs a semantic search using the prompt and provides top results for the best learning path.

One of the important feature of this system is its summarization module, where we shorten lengthy course descriptions and recommended material into brief and accurate overviews. This not only allows user to comprehend easily but also improves decision-making by quickly surfacing relevant course content. On top of this, the semantic filtering function helps to perform queries and domain-specific educational terminology by understanding the underlying meaning of words rather than simply matching prompt keywords.

We used Microsoft Learn's API which is a high-quality dataset and is constantly updated, allowing our recommender system to remain relevant and useful over time. These courses are modular, concise, and focus on improvement of professional skills, making it an ideal catalog for users seeking structured growth in STEM domains.

Apart from the technical implementation, our program provides recommendations in paginated format, has summarized course overview, and links to these courses, making it user-friendly.

6 Limitations and Future Work

Despite the effectiveness of our system, several limitations could be addressed in future work. The recommendation system primarily rely on TF-IDF vectorization for semantic understanding. TF-IDF captures keyword frequency patterns, but it will not understand the keywords context. Modern deep learning models like BERT or Sentence Transformers are bidirectional and and grasp context of the keywords in the prompt. As a result, our system may misinterpret user intent in more complex queries or fail to recommend semantically similar content if the vocabulary differs significantly.

Another key limitation lies in the quality and structure of the input course catalog. Microsoft Learn provides relatively well-structured course descriptions, but not all courses or modules contain comprehensive summaries or consistent formatting. This may result in variations in recommendation quality in case essential keywords or concepts are absent from the course metadata.

The current recommendation system is limited to Microsoft Learn. The courses available on the platform are domain specific and lack diversity in course content. Users looking for broader educational resources across multiple platforms like edX, Coursera and Udacity may find this system too narrow in scope.

In future works, we would like our system to use Distilbert which is a lite version of BERT for semantic understanding and additionally, incorporate user profiles, learning history, or preferences—important factors in building more personalized and adaptive learning paths. Using course catalogs from edx or cousera will increase data size or number of courses offered, so using distilbert embeddings and Faiss for searching top recommendations might be helpful to make recommendations faster.

7 Conclusion

In conclusion, our research focuses on a functional and user-friendly AI-based course recommendation engine that utilizes the Microsoft Learn catalog and NLP methods to offer a semantic search experience. The integration of summarization, filtering, and course clustering based on similarity threshold highlights how lightweight models can enhance educational content discovery.

Using Tf-IDF vectorization and cosine similarity provides a solid foundation for recommendation systems and can be enhanced by including transformer-based models for deeper semantic understanding, multilingual support, and cross-platform course catalogs for better recommendations or a personalized learning path. Additionally, integrating user feedback or learning behaviors could further improve recommendation accuracy and relevance.

The broader implication of this work suggests that multiple techniques can be integrated and extended to other open educational datasets, making learning more accessible and personalized. As learning platforms keep evolving to keep up with new technologies, customizable solutions like this one for learners can help learning easier and make learning journeys smoother.

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